TONER-1

260,032 parameters
int8 weights, one scale per row
Inference runs inside the PostScript interpreter

State of the art among language models that execute inside a print job.

SAMPLE OUTPUT

Unedited model output. No prompt: generation starts at the BOS token.

Once upon a time, there was a little girl named Lily. She loved to play with her toys. One day, she went to the park with her mom and dad. They were very happy.

At the park, they saw a big box. The boy wanted to play with it. He wanted to play with the box. He put the box on the box and put it in the box. Then, he put the box in the box. The box was very happy.

The boy was happy too. He put the box in the box. He put the box in the box. The box was not a box. The boy was happy. He learned that somet…

TOKEN STREAM

All 200 sampled tokens, one box per vocabulary piece. · marks a space, ¶ a newline.

·Once ·upon ·a ·time , ·there ·was ·a ·little ·g ir l ·named ·Lily . ·She
·lo ved ·to ·play ·with ·her ·to y s . ·One ·day , ·she ·w ent ·to ·the
·p ar k ·with ·her ·mom ·and ·d ad . ·They ·we re ·very ·happy . ¶ A t
·the ·p ar k , ·they ·saw ·a ·big ·b o x . ·The ·b o y ·want ed ·to
·play ·with ·it . ·He ·want ed ·to ·play ·with ·the ·b o x . ·He ·p ut
·the ·b o x ·on ·the ·b o x ·and ·p ut ·it ·in ·the ·b o x . ·The n ,
·he ·p ut ·the ·b o x ·in ·the ·b o x . ·The ·b o x ·was ·very ·happy .
¶ T he ·b o y ·was ·happy ·to o . ·He ·p ut ·the ·b o x ·in ·the ·b o x
. ·He ·p ut ·the ·b o x ·in ·the ·b o x . ·The ·b o x ·was ·n ot ·a ·b
o x . ·The ·b o y ·was ·happy . ·He ·l e ar n ed ·that ·s om et

HOW THIS PAGE WAS MADE

1 HOST
Sends the print job. Performs no arithmetic.

→

2 INTERPRETER
Parses the int8 weights, then runs the forward pass once per token.

→

3 PAGE
The same job typesets the story and calls showpage.

TOKENS	SECONDS	TOKENS / SEC	SEED	TEMPERATURE
200	**2.4**	**82.0**	**42** (unused)	**0** greedy

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
MODEL	stories260K printer-tuned (Llama-2 architecture) · dim 64 · 5 layers · 8 query / 4 KV heads · 512-token vocab
STOPPED BY	Token budget: STEPS = 200. The story was cut off mid-thought.

Host compute: 0 FLOPs. Inference cost includes toner.

Every word above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seed on a different interpreter tells a different story.